ANSWER KEY & MARKING SCHEME SET 1

Chapter 14: **Probability** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $\frac{5}{36}$	Favourable outcomes: (2,6), (3,5), (4,4), (5,3), (6,2); number $= 5$; $P = \frac{5}{36}$
2	**(d)** $\frac{1}{6}$	Favourable outcomes: (1,1), (1,2), (1,3), (2,1), (2,2), (3,1); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$
3	**(d)** 45	$\frac{x}{15+x} = 3 \times \frac{15}{15+x}$ gives $x = 3 \times 15$; $x = 45$
4	**(c)** 5	Cross-multiplying: $7(7+x) = 4(16+x)$; $x = 5$
5	**(d)** $\frac{4}{5}$	Favourable outcomes = number of red balls; $P = \frac{20}{25} = \frac{4}{5}$
6	**(a)** $\frac{19}{70}$	Favourable numbers: 19 outcomes; number $= 19$; $P = \frac{19}{70}$
7	**(d)** $\frac{43}{50}$	Favourable outcomes = number of good pens; $P = \frac{215}{250} = \frac{43}{50}$
8	**(b)** $\frac{1}{6}$	Favourable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$
9	**(a)** Both A and R are true and R is the correct explanation of A.	No face of a die shows more than 6, so the event is impossible and its probability is 0.
10	**(c)** A is true but R is false.	Primes are 2, 3, 5, so $P = \frac{3}{6} = \frac{1}{2}$: A true; probability lies between 0 and 1, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,3), (2,2), (2,6), (3,1), (3,5), (4,4), (5,3), (6,2), (6,6); number $= 9$ (1); $P = \frac{9}{36} = \frac{1}{4}$ (1).	1+1
12	$P(\text{blue}) = \frac{x}{18+x}$ and $P(\text{red}) = \frac{18}{18+x}$; $\frac{x}{18+x} = 3 \times \frac{18}{18+x}$ gives $x = 3 \times 18$ (1); $x = 54$ (1).	1+1
13	After adding, red balls $= 15 + 6 = 21$ and total balls $= 30$; Favourable outcomes = number of red balls (1); $P = \frac{21}{30} = \frac{7}{10}$ (1).	1+1
14	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: 15 pairs; number $= 15$ (1); $P = \frac{15}{36} = \frac{5}{12}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (3,6), (4,5), (5,4), (6,3); number $= 4$ (1); $P = \frac{4}{36} = \frac{1}{9}$ (1).	1+1+1
16	$\frac{9+x}{16+x} = \frac{2}{3}$ (1); Cross-multiplying: $3(9+x) = 2(16+x)$ (1); $x = 5$ (1).	1+1+1
17	Total possible outcomes $= 84$ (1); Favourable numbers: 2; number $= 1$ (1); $P = \frac{1}{84}$ (1).	1+1+1
	OR: Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (2,2), (2,4), (2,6), (4,2), (4,4), (4,6), (6,2), (6,4), (6,6); number $= 9$ (1); $P = \frac{9}{36} = \frac{1}{4}$ (1).	1+1+1
18	$\frac{13+x}{24+x} = \frac{16}{27}$ (1); Cross-multiplying: $27(13+x) = 16(24+x)$ (1); $x = 3$ (1).	1+1+1
	OR: Total possible outcomes $= 66$ (1); Favourable numbers: 47 outcomes; number $= 47$ (1); $P = \frac{47}{66}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $4 + 5 + 6 = 15$ (1). (ii) $P(\text{red}) = \frac{4}{15}$ (1). (iii) $P(\text{not green}) = \frac{9}{15} = \frac{3}{5}$ (1). (iv) $P(\text{blue or green}) = \frac{11}{15}$ (1); new total $= 20$, red $= 9$, $P = \frac{9}{20}$ (1).	1+1+1+2
20	Total outcomes $= 36$ (2); Favourable outcomes: 8; 9; 14 (2); Probabilities: $\frac{2}{9}$, $\frac{1}{4}$, $\frac{7}{18}$ (1).	2+2+1
	OR: $\frac{9+x}{30+x} = \frac{8}{15}$ (2); Cross-multiplying: $15(9+x) = 8(30+x)$ (2); $x = 15$ (1).	2+2+1

Mistakes to avoid:

- Writing a probability greater than 1 or negative; a probability always lies between 0 and 1.
- Using outcomes that are not equally likely (e.g. treating 'sum 7' and 'sum 12' as equally likely).
- Counting 1 as a prime number, or leaving out 2, the only even prime.
- Counting the overlap twice in 'A or B' questions.

- For two dice, using only 12 or 6 outcomes instead of 36, or counting (2,6) and (6,2) as the same outcome.
- Not changing the total number of outcomes when balls are removed or added, and not simplifying the fraction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 14: **Probability** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 5	Cross-multiplying: $25(11+x)=16(20+x)$; $x=5$
2	**(c)** $\frac{20}{27}$	Favourable outcomes = number of red balls; $P=\frac{20}{27}$
3	**(d)** $\frac{37}{80}$	Favourable numbers: 37 outcomes; number $=37$; $P=\frac{37}{80}$
4	**(c)** $\frac{17}{20}$	Favourable outcomes = number of good pens; $P=\frac{153}{180}=\frac{17}{20}$
5	**(a)** $\frac{1}{6}$	Favourable outcomes: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6); number $=6$; $P=\frac{6}{36}=\frac{1}{6}$
6	**(a)** $\frac{1}{4}$	Favourable outcomes: (1,1), (1,3), (1,5), (3,1), (3,3), (3,5), (5,1), (5,3), (5,5); number $=9$; $P=\frac{9}{36}=\frac{1}{4}$
7	**(d)** 70	$\frac{x}{14+x}=5\times\frac{14}{14+x}$ gives $x=5\times14$; $x=70$
8	**(c)** 7	Cross-multiplying: $25(5+x)=12(18+x)$; $x=7$
9	**(d)** A is false but R is true.	A is false since $P(E)\le1$; R states exactly that and is true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Outcomes HH, HT, TH, TT: three have at least one head, so $P=\frac{3}{4}$. R is true but it is not the reason for the value.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$P(\text{blue})=\frac{x}{16+x}$ and $P(\text{red})=\frac{16}{16+x}$; $\frac{x}{16+x}=5\times\frac{16}{16+x}$ gives $x=5\times16$ (1); $x=80$ (1).	1+1
12	After adding, red balls $=12+8=20$ and total balls $=28$; Favourable outcomes = number of red balls (1); $P=\frac{20}{28}=\frac{5}{7}$ (1).	1+1
13	Total number of outcomes $=6\times6=36$; Favourable outcomes: (6,6); number $=1$ (1); $P=\frac{1}{36}$ (1).	1+1
14	$P(\text{blue})=\frac{x}{20+x}$ and $P(\text{red})=\frac{20}{20+x}$; $\frac{x}{20+x}=2\times\frac{20}{20+x}$ gives $x=2\times20$ (1); $x=40$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total number of outcomes $=6\times6=36$ (1); Favourable outcomes: (1,6), (2,6), (3,6), (4,6), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6); number $=11$ (1); $P=\frac{11}{36}$ (1).	1+1+1
16	$\frac{17+x}{26+x}=\frac{8}{11}$ (1); Cross-multiplying: $11(17+x)=8(26+x)$ (1); $x=7$ (1).	1+1+1
17	Total possible outcomes $=96$ (1); Favourable numbers: 45 outcomes; number $=45$ (1); $P=\frac{45}{96}=\frac{15}{32}$ (1).	1+1+1
	OR: Total number of outcomes $=6\times6=36$ (1); Favourable outcomes: (1,2), (2,1), (5,6), (6,5); number $=4$ (1); $P=\frac{4}{36}=\frac{1}{9}$ (1).	1+1+1
18	$\frac{6+x}{20+x}=\frac{8}{15}$ (1); Cross-multiplying: $15(6+x)=8(20+x)$ (1); $x=10$ (1).	1+1+1
	OR: Total possible outcomes $=72$ (1); Favourable numbers: 18 outcomes; number $=18$ (1); $P=\frac{18}{72}=\frac{1}{4}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $6\times6=36$ (1). (ii) Sum 8: (2,6), (3,5), (4,4), (5,3), (6,2), so $P=\frac{5}{36}$ (1). (iii) Doublets: 6, so $P=\frac{6}{36}=\frac{1}{6}$ (1). (iv) Sum 2: 1 way, 3: 2, 5: 4, 7: 6, 11: 2; total 15 favourable (1); $P=\frac{15}{36}=\frac{5}{12}$ (1).	1+1+1+2
20	Total outcomes $=36$ (2); Favourable outcomes: 6; 4; 7 (2); Probabilities: $\frac{1}{6}$, $\frac{1}{9}$, $\frac{7}{36}$ (1).	2+2+1
	OR: $\frac{11+x}{36+x}=\frac{16}{41}$ (2); Cross-multiplying: $41(11+x)=16(36+x)$ (2); $x=5$ (1).	2+2+1

Mistakes to avoid:

- Writing a probability greater than 1 or negative; a probability always lies between 0 and 1.
- Using outcomes that are not equally likely (e.g. treating 'sum 7' and 'sum 12' as equally likely).
- Counting 1 as a prime number, or leaving out 2, the only even prime.
- Counting the overlap twice in 'A or B' questions.
- For two dice, using only 12 or 6 outcomes instead of 36, or counting (2,6) and (6,2) as the same outcome.
- Not changing the total number of outcomes when balls are removed or added, and not simplifying the fraction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 14: **Probability** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{17}{20}$	Favourable outcomes = number of good pens; $P = \frac{272}{320} = \frac{17}{20}$
2	**(a)** $\frac{1}{9}$	Favourable outcomes: (2,6), (3,4), (4,3), (6,2); number = 4; $P = \frac{4}{36} = \frac{1}{9}$
3	**(d)** $\frac{5}{12}$	Favourable outcomes: 15 pairs; number = 15; $P = \frac{15}{36} = \frac{5}{12}$
4	**(b)** 63	$\frac{x}{9+x} = 7 \times \frac{9}{9+x}$ gives $x = 7 \times 9$; $x = 63$
5	**(b)** 6	Cross-multiplying: $2(9 + x) = 1(24 + x)$; $x = 6$
6	**(a)** $\frac{10}{11}$	Favourable outcomes = number of red balls; $P = \frac{30}{33} = \frac{10}{11}$
7	**(d)** $\frac{23}{90}$	Favourable numbers: 23 outcomes; number = 23; $P = \frac{23}{90}$
8	**(c)** $\frac{87}{100}$	Favourable outcomes = number of good pens; $P = \frac{348}{400} = \frac{87}{100}$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$P(\text{not } E) = 1 - 0.35 = 0.65$ using R.
10	**(c)** A is true but R is false.	Favourable outcomes 3 out of 8 gives $\frac{3}{8}$: A true; probability uses *favourable* outcomes, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	After adding, red balls = $21 + 9 = 30$ and total balls = 39; Favourable outcomes = number of red balls (1); $P = \frac{30}{39} = \frac{10}{13}$ (1).	1+1
12	Total number of outcomes = $6 \times 6 = 36$; Favourable outcomes: (1,4), (2,3), (3,2), (4,1); number = 4 (1); $P = \frac{4}{36} = \frac{1}{9}$ (1).	1+1
13	$P(\text{blue}) = \frac{x}{12+x}$ and $P(\text{red}) = \frac{12}{12+x}$; $\frac{x}{12+x} = 7 \times \frac{12}{12+x}$ gives $x = 7 \times 12$ (1); $x = 84$ (1).	1+1
14	After adding, red balls = $10 + 6 = 16$ and total balls = 30; Favourable outcomes = number of red balls (1); $P = \frac{16}{30} = \frac{8}{15}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total number of outcomes = $6 \times 6 = 36$ (1); Favourable outcomes: 25 pairs; number = 25 (1); $P = \frac{25}{36}$ (1).	1+1+1
16	$\frac{10+x}{24+x} = \frac{9}{16}$ (1); Cross-multiplying: $16(10 + x) = 9(24 + x)$ (1); $x = 8$ (1).	1+1+1
17	Total possible outcomes = 54 (1); Favourable numbers: 47 outcomes; number = 47 (1); $P = \frac{47}{54}$ (1).	1+1+1
	OR: Total number of outcomes = $6 \times 6 = 36$ (1); Favourable outcomes: (1,6), (2,3), (3,2), (6,1); number = 4 (1); $P = \frac{4}{36} = \frac{1}{9}$ (1).	1+1+1
18	$\frac{15+x}{20+x} = \frac{6}{7}$ (1); Cross-multiplying: $7(15 + x) = 6(20 + x)$ (1); $x = 15$ (1).	1+1+1
	OR: Total possible outcomes = 90 (1); Favourable numbers: 24 outcomes; number = 24 (1); $P = \frac{24}{90} = \frac{4}{15}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 25 (1). (ii) Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23 (9 numbers), $P = \frac{9}{25}$ (1). (iii) Multiples of 5: 5, 10, 15, 20, 25 (5 numbers), $P = \frac{5}{25} = \frac{1}{5}$ (1). (iv) 5 is counted in both lists, so prime or multiple of $5 = 9 + 5 - 1 = 13$ numbers (1); $P(\text{neither}) = \frac{25-13}{25} = \frac{12}{25}$ (1).	1+1+1+2
20	Total outcomes = 36 (2); Favourable outcomes: 11; 4; 4 (2); Probabilities: $\frac{11}{36}$, $\frac{1}{9}$, $\frac{1}{9}$ (1).	2+2+1
	OR: $\frac{19+x}{20+x} = \frac{40}{41}$ (2); Cross-multiplying: $41(19 + x) = 40(20 + x)$ (2); $x = 21$ (1).	2+2+1

Mistakes to avoid:

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- Counting 1 as a prime number, or leaving out 2, the only even prime.
- Counting the overlap twice in 'A or B' questions.
- For two dice, using only 12 or 6 outcomes instead of 36, or counting (2,6) and (6,2) as the same outcome.

- Not changing the total number of outcomes when balls are removed or added, and not simplifying the fraction.

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